AI-Powered Intelligent Highway Entry Safety Screening System (AI-HESS)

Technical Progress Report

Team: NodeNova

Team Members: Weerasinghe W.A.N.D. , Jayakody G.K. , Wijesinghe W.G.S.S. , Dulanjana Yatagama

1. Project Overview

AI-HESS is an AI-powered highway entry safety screening system developed to identify potential vehicle and environmental risks before vehicles enter high-speed highways. The proposed system combines Artificial Intelligence, Computer Vision, IoT sensors, and environmental monitoring to assess vehicle safety and generate a safety score.

2. Technical Progress

Our team has successfully developed the initial simulation of the AI-HESS system, demonstrating the proposed highway-entry safety screening workflow. The simulation provides a foundation for developing the physical prototype.

The required hardware and equipment have been identified and collected for the prototype. Relevant datasets have also been identified and collected for developing the AI and computer-vision components.

Based on the collected datasets, AI model training is currently in progress for the proposed vehicle safety inspection system. The team is working on improving the models and preparing them for evaluation. In parallel, preparations are underway for integrating the collected hardware, sensors, and AI components into the physical prototype.

The overall technical workflow has been established, covering vehicle inspection, data collection, AI analysis, and safety assessment. The proposed architecture combines computer vision, deep learning, IoT sensing, and environmental monitoring.

3. Next Technical Steps

The next phase will focus on completing the initial AI models, evaluating their performance, integrating the collected hardware and sensors, and connecting the AI components with the physical prototype. The team will then conduct system-level testing and progressively integrate vehicle inspection, environmental monitoring, and safety-score generation.